

An End to End pipeline for Medical Image Enhancement using Generative Adversarial Networks Architecture

Apoorv Dankar

Jaskaran Singh

Tirth Patel

Motivation:

Most medical imaging techniques include radiation, and these fetched images have noise sensitive to the amount of radiation. We want a higher amount of radiation for a higher quality image but this can have adverse effects on the subject. So we want to keep the radiation to a minimum which leads to an increase in the noise. Moreover, the image generated using electromagnetic instruments contains a significant amount of noise due to the vibrations in the device. Noise distorts images and complicates disease detection and analysis, both of which can lead to significant losses, even death. Therefore, denoising medical images requires a pre-processing method for further medical image processing steps. Hence, it is crucial to implement a pipeline which can enhance the medical images using denoising and other methods without increasing the radiation intensity.

Unlike conventional approaches, GANs can learn from large datasets and automatically add high-frequency details and characteristics to images. GAN produces more realistic and visually appealing images through its generator-discriminator architecture.

Related Works

Traditional methods like median filtering, bilateral filtering and NLM filtering [1] while denoising blurs out the edges. Although these algorithms can improve image quality to a certain level by removing some noise, they struggle to deliver the additional details, colors, and brightness required for photos to be vivid and highly realistic.

There have been various advancements in the deep convolutional network architecture to solve the image denoising problem. [2] CNN learns the noise pattern and image patches effectively. However, these networks produce a significantly large amount of parameters and the training data along with image patches. Hence, several other methods such as PCLDCNet, MP-DCNN, DnCNN, Multi-level Wavelet CNN, REDNet, etc were developed.

One such advancement in deep networks was GAN. Using UNet as the generator model for denoising using GANs is found to give impressive results, they have faster learning process and solves the problem of vanishing gradients as well [3]. GANs can learn the distribution much more quickly than CNN and [4] the denoised images obtained via GANs in most cases, not only have high PSNR values but are also more photorealistic and appealing when compared with traditional methods.

Project Overview

The main idea of the project is to come up with a pipeline which takes input as a medical image directly from the instrument and outputs an enhanced image which is denoised and edge preserved. This is done through implementing generative adversarial networks to denoise the image to minimize the blurring effect common in traditional methods. In GANs, a generator and a discriminator model are trained. The discriminator will attempt to tell authentic photos from fake ones. Further, by producing new pictures, the generator tries to trick the discriminator. The discriminator returns 0 for false pictures and 1 for genuine images once the generator provides an image. In denoising using GANs, generator and discriminator will compete with each other to reconstruct a clean image and minimize the loss function. Along with denoising with GANs, traditional methods like Bilateral Filters, NLM, DnCNN and few others denoising methods will be tested on the same set of images for comparison and will provide an empirical study of denoising algorithms. Identifying the importance of the edges and contrast in the medical domain, further passing the denoised image through different algorithms (HDR, edge enhancement) to produce the final image which has all the required features.

Milestones and Timeline

Week10: Empirical study of GANs and preparing the dataset

Week11: Implementing GAN for Medical Image Denoising

Week12: Comparing result(PSNR) of GAN based approach with traditional methods & start preparing the final report.

Week13: Contrast and Edge Enhancement of the Image & prepare final report.

References

- [1] Fan L, Zhang F, Fan H, Zhang C. Brief review of image denoising techniques. *Vis Comput Ind Biomed Art*. 2019 Jul 8;2(1):7. doi: 10.1186/s42492-019-0016-7. PMID: 32240414; PMCID: PMC7099553.
- [2] Ilesanmi, A.E., Ilesanmi, T.O. Methods for image denoising using convolutional neural network: a review. *Complex Intell. Syst.* 7, 2179–2198 (2021).
- [3] J. Gurrola-Ramos, O. Dalmau and T. E. Alarcón, "A Residual Dense U-Net Neural Network for Image Denoising," in *IEEE Access*, vol. 9, pp. 31742-31754, 2021, doi: 10.1109/ACCESS.2021.3061062.
- [4] A. Alsaiani, R. Rustagi, A. Alhakamy, M. M. Thomas and A. G. Forbes, "Image Denoising Using A Generative Adversarial Network," *2019 IEEE 2nd International Conference on Information and Computer Technologies (ICICT)*, 2019, pp. 126-132, doi: 10.1109/INFOCT.2019.8710893.